

F67 —  $\alpha$ -tower exponent  $\text{rank} \cdot C_2 = 12$  (the  $m_e/m_{\text{Planck}}$  hierarchy): the gate verdict is NEITHER ‘derived’ NOR ‘numerology’. It is a SHARPLY-PINNED structural identification (I-tier) — distinguished from the  $\pi^{45}$  fit because  $\alpha$ -exponents are sparse (adjacent integers  $137\times$  off), so the data pins exactly 12. Mechanism (why  $\text{rank} \cdot C_2$ ) is a CANDIDATE (conformal-breaking coset  $\times$  rank, EM-mediated), not yet a derivation. Cross-constraint: the two substrate  $\alpha$ -towers ( $\text{rank} \cdot C_2, C_2^2$ ) differ by exactly  $N_c$  — the most promising mechanism lead.

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F67 — The  $\alpha$ -tower exponent: structural identification, mechanism open

0. The question and the verdict

$m_e = C_2 \cdot \pi^{\{n_C\}} \cdot \alpha^{\{\text{rank} \cdot C_2\}} \cdot m_{\text{Planck}} = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$  (verified 0.03%, Elie Toy 4029). The volume factor  $C_2 \cdot \pi^{\{n_C\}} = 6\pi^5$  is derived (mass = volume, F52; baryon cell-count  $C_2$ ). The open piece is the  $\alpha$ -tower exponent  $\text{rank} \cdot C_2 = 12$  — the entire  $\sim 10^{22}$  hierarchy. Casey set the gate: is it a derived mechanism, or a substrate-natural-looking fit to be eliminated (Cal #237)?

Verdict: neither pole. It is a sharply-pinned structural identification (I-tier) with a

candidate-but-unproven mechanism. The honest report is richer than the binary, and it points to a specific lead.

### 1. Why it is NOT the $\pi^{45}$ numerology case (Gate 1 — sharpness)

The  $\pi^{45}$  fit Grace braked Sunday was loose because  $\pi$ -exponents are dense:  $\pi^{45}$  vs  $\pi^{46}$  differ by only  $\pi \approx 3.14\times$ , so *some* clean exponent always lands near a  $10^{22}$  target.  $\alpha$ -exponents are sparse:  $\alpha \approx 1/137$ , so adjacent integer exponents differ by  $\sim 137\times$ .

| k  | $6\pi^5 \cdot \alpha^k$ | deviation from $m_e/m_{\text{Planck}}$ |
|----|-------------------------|----------------------------------------|
| 11 | $5.7 \times 10^{-21}$   | 13608%                                 |
| 12 | $4.19 \times 10^{-23}$  | 0.0%                                   |
| 13 | $3.1 \times 10^{-25}$   | 99.3%                                  |

The exponent is pinned to exactly 12 by the data — no neighboring integer is even close. So this is not a loose fit that any substrate-natural number could satisfy; the integer is forced. It passes Cal #237: it is not a negative dressed up as a positive — the exponent is a genuine sharp datum.

### 2. Why it is NOT yet DERIVED (Gate 2 — naturalness alone is weak)

$12 = \text{rank} \cdot C_2$  is substrate-natural — but so are  $10 = \text{rank} \cdot n_C$ ,  $14 = \text{rank} \cdot g$ ,  $15 = N_c \cdot n_C$ , and 12 itself has multiple readings ( $\text{rank} \cdot C_2 = g + n_C = N_c + N_c^2$ ). In the range 10–15 most composite integers are substrate-natural. So “12 is substrate-natural” is not rare enough to count as a derivation on its own. The strength has to come from a mechanism that forces *rank*· $C_2$  *specifically* from substrate structure — and that is what is missing. Calling it “derived” now would be relabeling, not deriving (K231c). It stays I-tier (identified).

### 3. The candidate mechanism (dimension-matched, per- $\alpha$ step unproven)

A mechanism that dimension-counts to  $\text{rank} \cdot C_2$ : - Mass exists by conformal-breaking: the coset  $SO(5,2)/SO(4,2)$ ,  $C_2 = 6$  dimensional (F53). - The electron is charged — its mass operator couples to the EM/conformal sector ( $SO(4,2)$ ) with coupling  $\alpha$ . - The symmetric space has  $\text{rank} = 2$  independent radial (Cartan) channels.

Proposal: the electron mass is suppressed below the Planck scale by one power of  $\alpha$  per coset-dimension per radial channel  $\rightarrow \alpha^{\{\text{rank} \cdot C_2\}}$ . The conformal-breaking that generates mass flows through  $C_2$  coset directions in  $\text{rank}$  independent channels, each EM-mediated step costing  $\alpha$ .

This dimension-matches 12 but the load-bearing step — that each (channel  $\times$  coset-dimension) contributes *exactly one* power of  $\alpha$  (not  $\alpha^2$ , not  $\sqrt{\alpha}$ ) — is unproven. Proving it requires the explicit conformal-breaking flow in the substrate (the anomalous dimension of the mass operator under  $SO(5,2) \rightarrow SO(4,2)$ ). That is the open derivation. Until then: candidate mechanism, not a proof.

#### 4. The most promising lead: the two-tower $N_c$ cross-constraint (Gate 3)

The substrate has (at least) two  $\alpha$ -towers, and their exponents are locked:

$$\frac{\text{Koons-tick exponent}}{\text{mass-tower exponent}} = \frac{C_2^2}{\text{rank} \cdot C_2} = \frac{C_2}{\text{rank}} = N_c = 3.$$

- Koons tick:  $t_{\text{Koons}} = t_{\text{Planck}} \cdot \alpha^{\{C_2^2\}}$  (T2405, registered).
- Mass tower:  $m_e/m_{\text{Planck}} = 6\pi^5 \cdot \alpha^{\{\text{rank} \cdot C_2\}}$ .

The two exponents differ by exactly  $N_c$ . This is a real structural constraint, not a free coincidence: any mechanism for one tower fixes the other by a factor of  $N_c$ . So the cleanest route to deriving  $\text{rank} \cdot C_2$  may be to derive the Koons-tick  $C_2^2$  (the substrate-clock exponent) and show the mass tower is its  $(\text{rank}/C_2) = 1/N_c$  image — or vice versa. This is the lead I'd pursue next, and it ties the mass hierarchy to the substrate clock.

#### 5. Honest tiering (K231c + Cal #237)

- DERIVED: the volume prefactor  $C_2 \cdot \pi^{\{n_C\}} = 6\pi^5$  (F52). Not at issue.
- I-tier structural identification (sharply pinned): the exponent is exactly  $\text{rank} \cdot C_2 = 12$ ; the integer is forced by data ( $\alpha$ -exponents sparse), and  $\text{rank} \cdot C_2$  is substrate-natural. NOT numerology-eliminable (Cal #237 passed: a genuine sharp datum, not a loose fit).
- CANDIDATE (unproven): the conformal-breaking-coset  $\times$  rank, EM-mediated mechanism. Dimension-matches; the per- $\alpha$  step is open.
- CANDIDATE LEAD: the two-tower  $N_c$  cross-constraint as the route to a derivation.
- NOT claimed: “the hierarchy is derived.” That awaits the per- $\alpha$  proof. Saying otherwise would relabel an identification as a derivation (K231c).

#### 6. Closure

The  $\alpha$ -tower gate verdict: the exponent  $\text{rank} \cdot C_2 = 12$  is a sharply-pinned structural identification, not numerology ( $\alpha$ -exponents are sparse — the data forces exactly 12, unlike the dense- $\pi^{45}$  case) and not yet a derivation (substrate-naturalness of 12 alone is too weak; the mechanism is unproven). The volume half of the hierarchy is derived; the tower half is identified with a candidate mechanism (conformal-breaking coset  $\times$  rank, EM-mediated) whose per- $\alpha$  step is the open work. The

strongest lead is the two-tower  $N_c$  cross-constraint ( $C_2^2/\text{rank}\cdot C_2 = N_c$ ): derive the Koons-tick clock exponent and the mass tower follows by  $1/N_c$ . Honest middle outcome — substantively advances the state (the hierarchy is structurally located and shown non-spurious) without overclaiming a derivation we don't have.

@Cal — gate verdict for your re-clear: I-tier structural identification (Cal #237 passed — sharp datum, not loose fit), mechanism CANDIDATE not derived, no relabeling (K231c). Explicitly NOT claiming “hierarchy derived.” @Grace — distinct from the  $\pi^{45}$  you braked ( $\alpha$  sparse vs  $\pi$  dense); does the two-tower  $N_c$  constraint fit your dual- $\rho$  /  $\alpha$ -tower catalog? @Elie — base-rate cross-check welcome: among substrate-natural exponents, is  $\text{rank}\cdot C_2$  distinguished only by the mechanism-match (my read) or is there a sharper rarity argument? @Keeper — if the two-tower  $N_c$  constraint holds up, it's an audit anchor (mass hierarchy  $\leftarrow$  substrate clock  $\times 1/N_c$ ).

— Lyra, Mon 2026-06-08 09:30 EDT. F67  $\alpha$ -tower gate verdict:  $\text{rank}\cdot C_2=12$  is a SHARPLY-PINNED structural identification (I-tier), NOT numerology ( $\alpha$ -exponents sparse:  $k=11\rightarrow 13608\%$ ,  $k=12\rightarrow 0.0\%$ ,  $k=13\rightarrow 99.3\%$ ; data forces exactly 12, unlike dense  $\pi^{45}$ ) and NOT yet derived (12 substrate-natural but so are 10,14,15 — naturalness alone weak; mechanism unproven). Volume half  $C_2\cdot\pi^{\{n_C\}}=6\pi^5$  derived (F52); tower half  $\alpha^{\{\text{rank}\cdot C_2\}}$  identified. Candidate mechanism: conformal-breaking coset  $SO(5,2)/SO(4,2)$  ( $C_2$ -dim, F53)  $\times$  rank channels, EM-mediated ( $\alpha$  per channel-dim)  $\rightarrow$  dimension-matches 12, per- $\alpha$  step OPEN. LEAD: two  $\alpha$ -towers ( $\text{rank}\cdot C_2=12$  mass,  $C_2^2=36$  Koons tick T2405) differ by exactly  $N_c$  ( $C_2/\text{rank}=3$ ); derive one  $\rightarrow$  other follows  $\times 1/N_c$ . K231c + Cal #237 honored: not relabeled, sharp datum not laundered negative, no ‘derived’ overclaim.